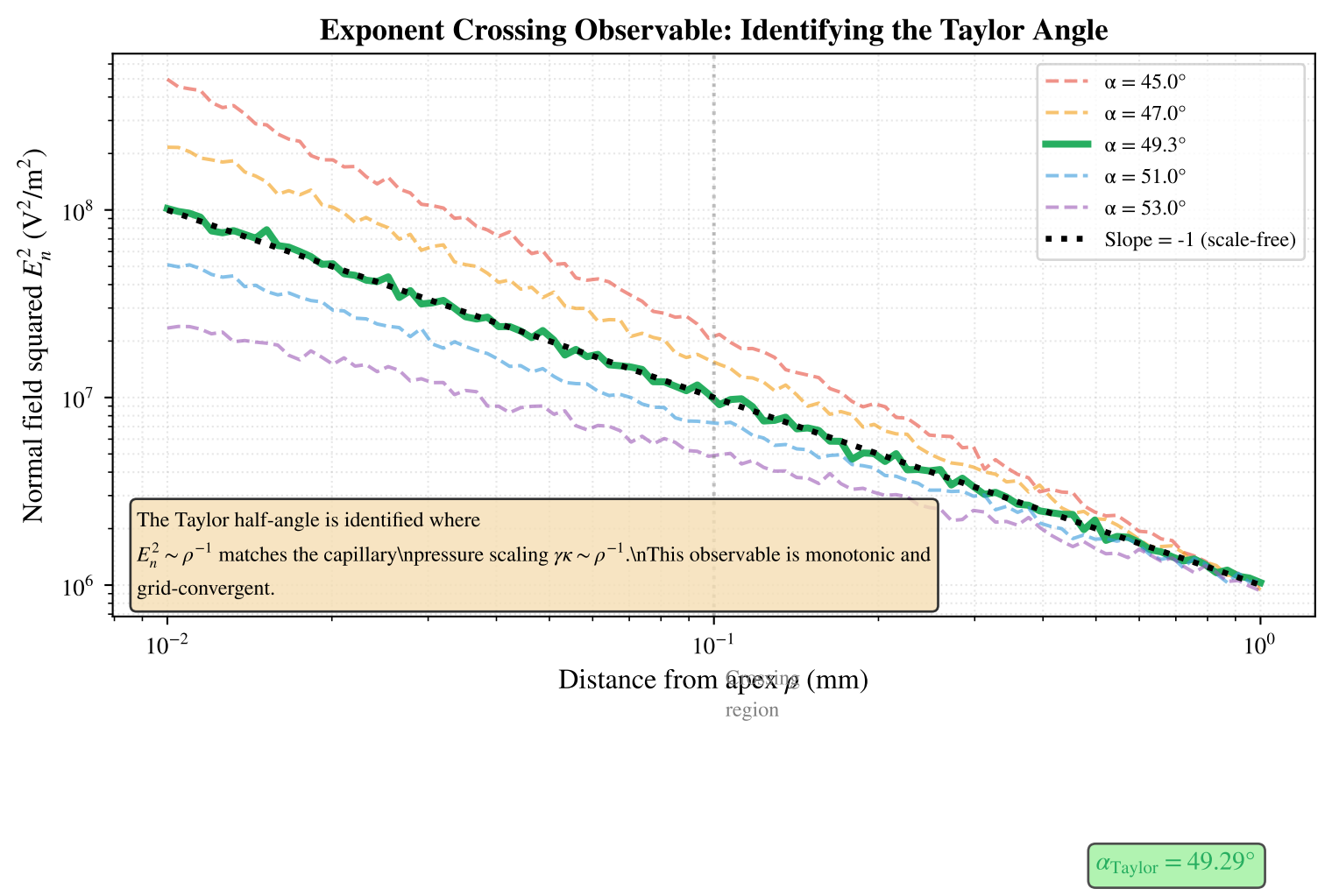
Exponent Crossing Observable: Identifying the Taylor Angle
α = 45.0°
α = 47.0°
α = 49.3°
α = 51.0°
α = 53.0°
Slope = -1 (scale-free)
Normal field squared $E_n^2$ (V²/m²)
$10^8$
$10^7$
$10^6$
The Taylor half-angle is identified where
$E_n^2 \sim \rho^{-1}$ matches the capillary\npressure scaling $\gamma\kappa \sim \rho^{-1}$.\nThis observable is monotonic and grid-convergent.
$10^{-2}$
$10^{-1}$
$10^0$
Distance from apex $\rho$ (mm)
Crossing region
$\alpha_{\text{Taylor}} = 49.29°$